

Living near the tracks: does station proximity reduce car ownership, and does rail reliability matter?

TIL6022 Python Programming (Project Proposal)

Group: 19

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Author contribution statement

All authors contributed equally to formulating the research questions, selecting and checking the datasets, and writing this proposal.

Introduction

Car ownership is an important aspect of mobility and is closely related to car dependency. While cars provide travellers with accessibility and flexibility, high levels of car dependency are also associated with societal and environmental impacts. These include energy consumption, impacts on public health and the use of urban space (Muñoz et al., 2024). In the Netherlands, car ownership varies across regions. Private car ownership per resident has decreased in highly urbanised areas, while it has increased in more rural areas (Zijlstra et al., 2022). These differences are related to a combination of factors, including differences in travel distances and the availability of alternatives to private car use (Zijlstra et al., 2022). Previous research shows that car ownership is associated with both socio-economic and spatial characteristics, including income, urban density and access to public transport (Feyzollahi et al., 2024).

Public transport accessibility is one of the spatial factors that can be related to car ownership. Previous research generally shows that better access to public transport is associated with lower car ownership, although this relationship varies between locations and population groups (Feyzollahi et al., 2024). However, living close to public transport does not necessarily mean that households own fewer cars. Shen et al. (2016), found that proximity to a metro station was associated with higher metro use, but not with lower car ownership. Similarly, Chatman (2013) found lower car ownership and car use among households living near railway stations, but showed that factors such as parking availability, housing characteristics, density and bus services also played an important role. In the Netherlands, Debrezion (2009) found that station choice depends not only on distance, but also on station characteristics such as service frequency and facilities. This suggests that the relationship between railway accessibility and car ownership may also depend on the quality of the rail service provided.

An important aspect of public transport service quality is reliability. Unreliable services can lead to longer or unpredictable waiting times and missed connections, which can make public transport less attractive to travellers. Chakrabarti (2015) found that higher service reliability was associated with higher public transport use. This suggests that having a station nearby may be less valuable when the service provided is frequently disrupted. However, while reliability has been studied in relation to public transport use, its role in the relationship between station proximity and car ownership is less clear.

This study therefore examines the relationship between railway station proximity and car ownership across neighbourhoods in the Netherlands. It examines whether neighbourhoods located closer to a railway station have lower car ownership, while accounting for socio-economic and spatial differences between neighbourhoods. In addition, station type and railway disruptions are considered to examine whether this relationship differs depending on the characteristics and reliability of the nearby station. The study aims to provide a better

understanding of how station proximity, station characteristics and railway reliability are related to car ownership in the Netherlands.

Research objective

The project aims to quantify the link between station proximity and car ownership across all Dutch neighbourhoods, and to test whether rail reliability moderates it.

1. Construct a reproducible neighbourhood-level dataset linking 2023 CBS data to Dutch railway stations and their disruption exposure over 2021–2023.
2. Describe and map the spatial distribution of household car ownership and station proximity across Dutch neighbourhoods in 2023, together with railway disruption exposure over 2021–2023.
3. Estimate the relationship between station distance and car ownership based on 2023 CBS neighbourhood data, controlling for income, urbanisation, household size and population density.
4. Test whether the relationship between station distance and car ownership in 2023 differs by station type and by station-level disruption exposure over 2021–2023.
5. Use the findings to discuss the implications of station proximity, station type and rail reliability for housing development near railway stations.

Research questions

Main question: Is there an association between distance to the nearest railway station and household car ownership in Dutch neighbourhoods (2023), and to what extent does this association vary by station type and disruption frequency (2021–2023)?

SQ1: How were average car ownership per household, distance to the nearest station, and train disruptions spread across Dutch neighbourhoods in 2023?

SQ2: What was the baseline between distance to the nearest station and car ownership per household in 2023, when accounting for income, urbanisation, household size, and density?

SQ3: How did the association between station distance and car ownership per household differ between major stations (intercity/junction hubs) and minor local stations in 2023?

SQ4: Was the association between living near a station and car ownership per household weaker in neighbourhoods near stations with frequent train disruptions (2021–2023)?

Geographical and temporal scale

- **Geographical scale:** all Dutch neighbourhoods as defined by CBS in 2023, linked to all Dutch railway stations.
- **Temporal scale:** a cross-sectional analysis with 2023 as reference year. We use the 2023 data so neighbourhood codes match. Disruption exposure is averaged over 2021–2023 to smooth it out. We thought that otherwise you do not have enough data if you use just one year. One outlier will then effect your whole outcome.

Datasets

Dataset	Source	Key variables
Kerncijfers wijken en buurten 2023 (85618NED)	CBS	Cars per household, income, urbanisation, household size
Nabijheid voorzieningen 2023 (85830NED)	CBS	Road distance to nearest train station and major transfer station

Dataset	Source	Key variables
Wijk- en buurtkaart 2023	CBS / Kadaster (PDOK)	Neighbourhood polygons (EPSG:28992)
Railway stations	Rijden de Treinen	Station code, station type, coordinates
Train disruptions 2021–2023	Rijden de Treinen	Affected stations, cause group, duration (minutes)

Intended data analysis pipeline

The data pipeline constructs a single, integrated dataset at the neighborhood level. This will directly serve both spatial visualisations and regression models.

Phase 1: Data acquisition and centralized data storage

The analysis begins by gathering all required primary datasets: national neighborhood statistics and accessibility indicators from CBS (*Kerncijfers wijken en buurten 2023*, *Nabijheid voorzieningen; afstand locatie, wijk- en buurtcijfers 2023*, *Toelichting variabelen Kerncijfers wijken en buurten 2024*), spatial boundary maps (*Wijk- en buurtkaart 2023, versie 3*), and historical station and disruption records from Rijden de Treinen (*Open data: train disruptions*). All raw data files are placed in a storage folder to guarantee good reproducibility and preserve the original baseline datasets.

Phase 2: Neighborhood sample filtering and quality control

To establish a representative analytical sample, the combined CBS datasets and their data are systematically filtered. The observations focus strictly on neighborhood-level records. Non-residential areas (entries with missing car ownership values) and statistical outliers (commercial zones where corporate leasing fleets artificially inflate car counts) are removed from the dataset.

Phase 3: Spatial matching of neighborhood and station attributes

By using spatial analysis, each neighborhood's geographic center is identified and paired with its nearest railway station. This process will add station characteristics to each neighborhood entry. This includes the station's identification code, location coordinates, and service classification (such as major transfer hubs versus local stations).

Phase 4: Station reliability quantification

The multi-year disruption logs are processed to quantify historical service dependability across the rail network. By evaluating the disruption records over a three-year period, we generate standardized performance metrics for each station. It specifically calculates the average annual number of service interruptions and total delay minutes.

Phase 5: Master dataset integration

The cleaned neighborhood attributes, nearest-station details and station reliability metrics are merged using unique station identifiers. This step creates a unified master analysis table that brings together all the target metrics (car ownership), key station indicators (distance, station type and disruption exposure), and relevant demographic and socioeconomic controls (income levels, urbanisation rate, household size, and density).

Neighbourhood	Cars per household	Distance to station (km)	Nearest station (station code)	Station type	Avg. disruption minutes per year	Household income	Urbanisation	Household size	Population density
A	0.8	0.6	STA	Major	1850	36.500	High	5	7.200

Phase 6: Spatial and exploratory data analysis

Initial insights are developed through textual summaries and geographic visualisations. National maps that display patterns of car ownership and station proximity, are produced next to diagnostic scatter plots. These provide an intuitive baseline understanding of spatial transit accessibility and car reliance across Dutch neighborhoods.

Phase 7: Econometric and interaction modeling

To isolate and evaluate the underlying relationships between rail accessibility and vehicle ownership, statistical regression models are used. Underlying relationships could be:

- **Baseline relationship:** A multiple linear regression estimates how car ownership varies with distance to the nearest station while controlling for confounding factors like income, urbanisation, household size, and density.
- **Station hierarchy effects:** Interaction terms evaluate whether proximity to major intercity hubs has a stronger association with car ownership reduction, compared to smaller regional stations.
- **Disruption sensitivity:** Interaction terms test whether frequent train disruptions become less than the expected reduction in household car ownership near stations.

Phase 8: Policy synthesis and planning implications

The statistical outcomes and spatial maps are combined to get practical recommendations for urban development and transit policy. The findings will eventually offer evidence on whether station-area housing strategies effectively restrain car reliance on their own, or if their success relies on parallel investments in rail network reliability.

Data availability check

The cells below load each dataset to confirm that the data is accessible and contains the variables we need.

```
from pathlib import Path
import pandas as pd

raw_dir = Path('data/raw')

bestanden = ['kwb_2023.csv', 'nabijheid_2023.csv', 'wijkenbuurten_2023_v3.gpkg',
             'stations-2023-09-nl.csv',
             'disruptions-2021.csv', 'disruptions-2022.csv', 'disruptions-2023.csv']

for f in bestanden:
    print(f'{f:35s}', 'OK' if (raw_dir / f).exists() else 'ONTBREEKT')
```

kwb_2023.csv	OK
nabijheid_2023.csv	OK
wijkenbuurten_2023_v3.gpkg	OK
stations-2023-09-nl.csv	OK
disruptions-2021.csv	OK
disruptions-2022.csv	OK
disruptions-2023.csv	OK

```
kwb = pd.read_csv(raw_dir / 'kwb_2023.csv', low_memory=False)
print(kwb.shape)
kwb[['Coding_3', 'SoortRegio_2', 'PersonenautoSPerHuishouden_112',
     'GemiddeldInkomenPerInwoner_81', 'MateVanStedelijkheid_125']].head()
```

(18116, 128)

	Codering_3	SoortRegio_2	PersonenautoSPerH uishouden_112	GemiddeldInkomenP erInwoner_81	MateVanStedelijkh eid_125
0	NL00	Land	1.1	32.8	2.0
1	GM1680	Gemeente	1.4	32.8	5.0
2	WK168000	Wijk	1.3	34.1	5.0
3	BU16800000	Buurt	1.3	33.4	5.0
4	BU16800009	Buurt	1.5	49.5	5.0

```
nabijheid = pd.read_csv(raw_dir / 'nabijheid_2023.csv', low_memory=False)
print(nabijheid.shape)
nabijheid[['Codering_3', 'AfstandTotTreinstationsTotaal_90',
            'AfstandTotBelangrijkOverstapstation_91', 'AfstandTotOpritHoofdverkeersweg_89']].head()
```

(18116, 116)

	Codering_3	AfstandTotTreinstatio nsTotaal_90	AfstandTotBelangrijkO verstapstation_91	AfstandTotOpritHoofd verkeersweg_89
0	NL00	5.3	10.8	1.9
1	GM1680	12.4	14.4	1.5
2	WK168000	13.3	14.3	1.5
3	BU16800000	13.2	14.2	1.5
4	BU16800009	13.7	15.3	2.6

```
stations = pd.read_csv(raw_dir / 'stations-2023-09-nl.csv')
print(stations.shape)
stations[['code', 'name_long', 'type', 'geo_lat', 'geo_lng']].head()
```

(397, 11)

	code	name_long	type	geo_lat	geo_lng
0	HT	's-Hertogenbosch	knooppuntIntercit ystation	51.690480	5.293620
1	HTO	's-Hertogenbosch Oost	stoptreinstation	51.700554	5.318333
2	HDE	't Harde	stoptreinstation	52.409168	5.893611
3	ATN	Aalten	stoptreinstation	51.921327	6.578627
4	AC	Abcoude	stoptreinstation	52.278500	4.977000

```
storingen = pd.concat(
    [pd.read_csv(raw_dir / f'disruptions-{jaar}.csv') for jaar in [2021, 2022, 2023]],
    ignore_index=True,
)
print(storingen.shape)
storingen[['rdt_station_codes', 'cause_group', 'start_time', 'duration_minutes']].head()
```

(15541, 14)

	rdt_station_codes	cause_group	start_time	duration_minutes
0	HGL, HGL0, ODZ	rolling stock	2021-01-01 08:12:55	197.0
1	ASD, ASDM, ASSP, DMN, WP	rolling stock	2021-01-01 10:17:19	12.0
2	BRN, DLD, SD, ST, STZ	infrastructure	2021-01-01 10:44:22	421.0
3	GBG, HDB, MRB	rolling stock	2021-01-01 12:25:19	20.0
4	BGN, RB	accidents	2021-01-01 15:17:02	42.0

```
import geopandas as gpd

buurten_kaart = gpd.read_file(raw_dir / 'wijkenbuurten_2023_v3.gpkg', layer='buurten', rows=5)
buurten_kaart[['buurtcode', 'personenautos_per_huishouden',
                'treinstation_gemiddelde_afstand_in_km', 'geometry']].head()
```

c:\Users\alecm\anaconda3\envs\TIL6022-26\Lib\site-packages\pyogrio\core.py:34: RuntimeWarning: Could not detect GDAL data files. Set GDAL_DATA environment variable to the correct path.
_init_gdal_data()

	buurtcode	personenautos_per_huisshouden	treinstation_gemiddelde_afstand_in_km	geometry
0	BU09989999	-99997.0	-99997.0	MULTIPOLYGON (((123629.78 379674.57, 123627.21...
1	BU00349997	-99997.0	-99997.0	MULTIPOLYGON (((150087.299 479382.379, 150000....
2	BU00509997	-99997.0	-99997.0	MULTIPOLYGON (((155047.32 474836.246, 155049.0...
3	BU00609998	-99997.0	-99997.0	MULTIPOLYGON (((196000 608000, 195250 607500, ...
4	BU00729998	-99997.0	-99997.0	MULTIPOLYGON (((158000 581000, 158331.38 58046...

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